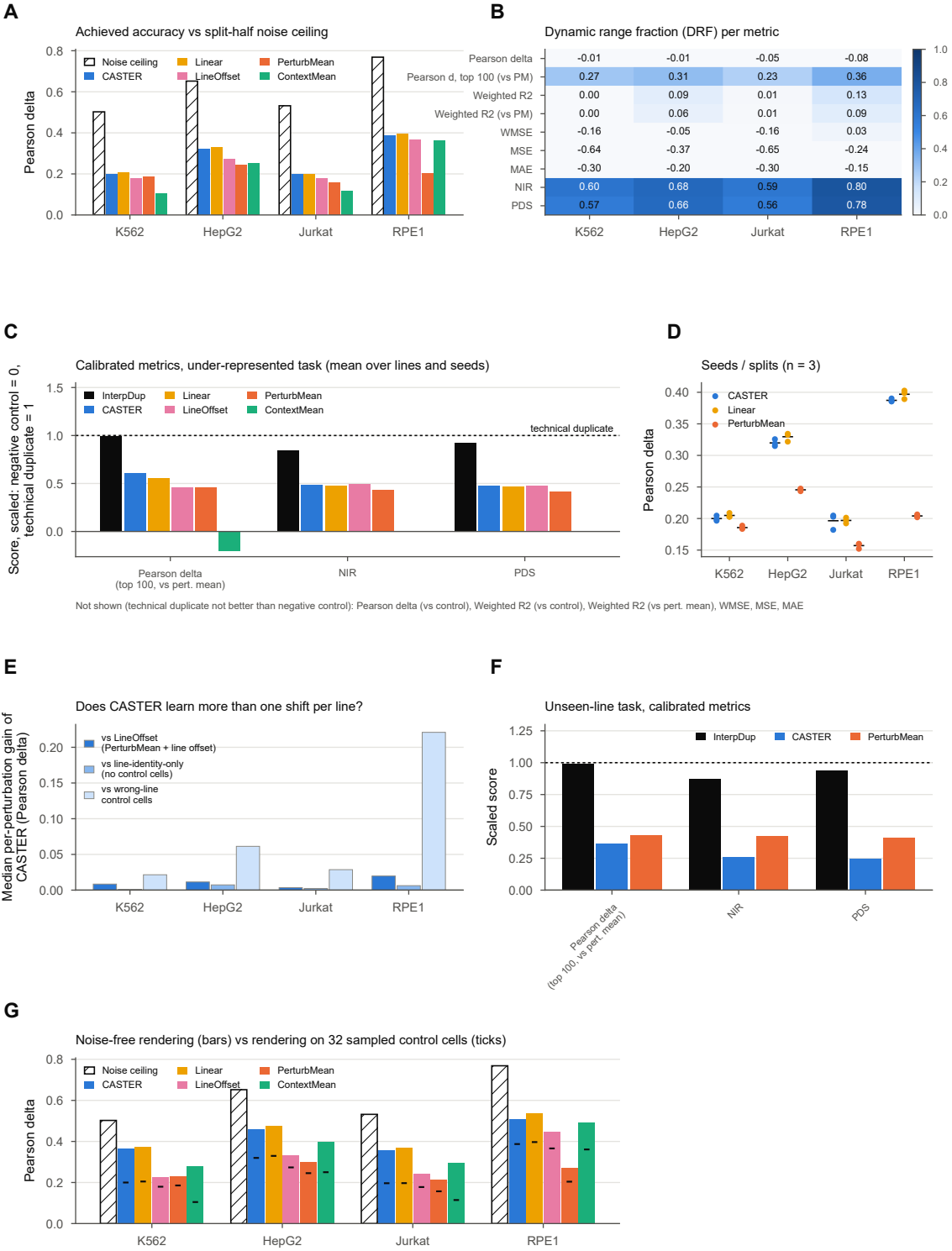

A
Achieved accuracy vs split-half noise ceiling
Pearson delta
Noise ceiling
CASTER
Linear
LineOffset
PerturbMean
ContextMean
K562
HepG2
Jurkat
RPE1
B
Dynamic range fraction (DRF) per metric
Pearson delta -0.01 -0.01 -0.05 -0.08
Pearson d, top 100 (vs PM) 0.27 0.31 0.23 0.36
Weighted R2 0.00 0.09 0.01 0.13
Weighted R2 (vs PM) 0.00 0.06 0.01 0.09
WMSE -0.16 -0.05 -0.16 0.03
MSE -0.64 -0.37 -0.65 -0.24
MAE -0.30 -0.20 -0.30 -0.15
NIR 0.60 0.68 0.59 0.80
PDS 0.57 0.66 0.56 0.78
C
Calibrated metrics, under-represented task (mean over lines and seeds)
Score, scaled: negative control = 0, technical duplicate = 1
InterpDup
technical duplicate
Pearson delta (top 100, vs pert. mean)
NIR
PDS
D
Seeds / splits (n = 3)
Pearson delta
Not shown (technical duplicate not better than negative control): Pearson delta (vs control), Weighted R2 (vs control), Weighted R2 (vs pert. mean), WMSE, MSE, MAE
E
Does CASTER learn more than one shift per line?
Median per-perturbation gain of CASTER (Pearson delta)
vs LineOffset (PerturbMean + line offset)
vs line-identity-only (no control cells)
vs wrong-line control cells
F
Unseen-line task, calibrated metrics
Scaled score
G
Noise-free rendering (bars) vs rendering on 32 sampled control cells (ticks)